PG No. 24 D_6 622 [hexagonal]

* character table

D_6	$1(1)$	$2_{001}(1)$	$2_{100}(3)$	$2_{120}(3)$	$3^+_{001}(2)$	$6^+_{001}(2)$
A_1	1	1	1	1	1	1
A_2	1	1	−1	−1	1	1
B_1	1	−1	−1	1	1	−1
B_2	1	−1	1	−1	1	−1
E_1	2	−2	0	0	−1	1
E_2	2	2	0	0	−1	−1

* polar ↔ axial conversion

A_1 (A_1)　A_2 (A_2)　B_1 (B_1)　B_2 (B_2)　E_1 (E_1)　E_2 (E_2)

* symmetric product

	A_1	A_2	B_1	B_2	E_1	E_2
A_1	A_1	A_2	B_1	B_2	E_1	E_2
A_2		A_1	B_2	B_1	E_1	E_2
B_1			A_1	A_2	E_2	E_1
B_2				A_1	E_2	E_1
E_1					$A_1 + E_2$	$B_1 + B_2 + E_1$
E_2						$A_1 + E_2$

* anti-symmetric product

A_1	A_2	B_1	B_2	E_1	E_2
−	−	−	−	A_2	A_2